

AAYUSH KUMAR

Software Engineer — Full Stack Developer

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Summary

Software Engineering student with experience building full-stack applications and scalable backend systems using React, Node.js, Express.js, MongoDB, and PostgreSQL. Passionate about backend engineering, REST APIs, performance optimization, and writing clean, maintainable code.

Skills

Languages: C++, JavaScript, Python, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Databases: MongoDB, PostgreSQL, Redis, Supabase Storage

Libraries: Socket.IO, Sharp

Tools: Git, GitHub, Postman, VS Code

Core: Data Structures & Algorithms, OOP, DBMS, REST API Design

Experience

Software Developer Intern

Feb 2026 – Present

Nexera

Remote

- Contributed to the development of a scalable full-stack learning platform using **React.js, Next.js, Node.js, Express.js, and PostgreSQL**.
- Implemented secure **Google OAuth authentication** and session management to provide seamless user login and access control.
- Developed backend APIs and frontend workflows for the **course doubt management system**, enabling efficient interaction between students and mentors.
- Integrated **Redis** to implement a real-time course like counter, improving responsiveness and reducing repeated database operations.
- Built responsive UI components and developed **admin functionalities** for managing platform content and user operations.

Projects

PixelFlow | *React.js, Express.js, Redis, Sharp*

[GitHub](#) [Live Demo](#)

- Built an adaptive image delivery platform with real-time optimization through intelligent format negotiation, responsive resizing, and on-demand transformations.
- Designed a high-performance delivery pipeline using **Sharp, Redis**, and object storage to generate, cache, and serve optimized image variants with low latency.
- Implemented secure REST APIs with **JWT**, Google OAuth 2.0, API key authentication, rate limiting, and input validation for user and third-party integrations.
- Built a React dashboard supporting batch uploads, analytics, folder management, image search, and real-time upload progress.

HomeHive | *Node.js, Express.js, PostgreSQL, Prisma, Socket.IO*

[GitHub](#) [Live Demo](#)

- Built a full-stack property marketplace supporting secure buyer-owner workflows with role-based access control, Google OAuth 2.0, and email/password authentication.
- Implemented encrypted real-time messaging using **Socket.IO** with RSA-OAEP encryption, enabling secure communication between buyers and property owners.
- Designed a cloud-based media pipeline using **Cloudinary**, delivering optimized property images with automatic compression, responsive **srcset** generation, and efficient asset management.
- Developed backend services using **PostgreSQL** and **Prisma**, including a custom session store, geospatial search, and buy-request workflows.

Education

B.Tech in Computer Science Engineering

Expected Graduation: 2028

Maharaja Agrasen Institute of Technology

Current CGPA: 8.28/10